

The Intersection of Computation, Economics, and Liberty: A Briefing Document

Executive Summary

The provided sources outline an emerging paradigm that synthesizes Hayekian market-process economics with computer science. This "Agoric" perspective views computation and software development not merely as technical disciplines, but as complex spontaneous orders governed by principles of property rights, decentralization, and freedom of expression. **Critical**

Takeaways:

- **Software as Speech:** A central argument posits that computer programs are a form of writing. Consequently, software patents are viewed as a violation of First Amendment rights and a form of government censorship.
- **Agoric Systems:** Research into "Agoric Open Systems" proposes using market mechanisms (such as competitive bidding for processor time and memory) to solve problems of resource allocation and complexity within large-scale computational networks.
- **The Digital Economy:** The development of anonymous digital cash and offshore electronic banking is identified as a primary catalyst for reducing state power to tax and regulate, while increasing individual privacy and monetary competition.
- **Spontaneous Order in AI:** Modern approaches to Artificial Intelligence (Emergent AI) mirror Hayekian theories of the mind, favoring decentralized, learning-based systems over centrally planned, rule-based "Expert Systems."
- **Transhumanist Foresight:** These technical and economic shifts are framed within a broader "Extropian" or transhumanist philosophy, predicting a future of radical technological breakthroughs in life extension, nanotechnology, and intelligence augmentation.

1. Software as Protected Expression

A foundational theme across the sources is the categorization of computer code as a form of writing, and therefore, protected speech.

The Constitutional Argument

Phil Salin argues that software is a literary endeavor requiring the same "skill, knowledge, invention, and creativity" as writing an essay.

- **Literacy Analogies:** Programming is compared to music composition or prose; though the code is "run" rather than "read" by the end-user, the underlying source code remains a form of human expression.
- **Censorship via Patent:** Software patents are characterized as government-granted monopolies over ideas. Salin asserts that patent examiners act as "de facto censors," violating the First Amendment by licensing what a writer may or may not publish.

Legal and Regulatory Challenges

The documents identify several historical points of friction between software and the state:

- **Patent Reform:** Critics argue that the 1981 *Diamond v. Diehr* Supreme Court decision was misread by administrative bodies to allow the patenting of software, a trend they seek to reverse.
- **Cryptography Export Controls:** The government has historically regulated encryption software as a "tool of war." Cases like the *Bernstein* federal court decision (1996) are cited as victories for the view of code as protected speech.
- **Regulatory Fiction:** The source notes that the National Security Agency (NSA) once deemed it illegal to publish the "Apache" server on the Web because it contained hooks for encryption, despite the illegality of such hooks being a "fictional creation" not found in actual regulations.

2. Agoric Systems: Markets in the Machine

"Agorics" refers to the application of market principles to computational resource management. This approach treats computational elements as economic agents.

Key Concepts in Agoric Computing

- **Software as Capital:** Software is described as a unique form of capital—pure knowledge that is reproducible at negligible cost and does not wear out with use.
- **Property Rights (OOPS):** Object-Oriented Programming Systems (OOPS) are viewed as a technical implementation of property rights. By "encapsulating" data and algorithms, these systems create boundaries that prevent one part of a program from interfering with another, mirroring the "tragedy of the commons" solutions found in human societies.
- **Competitive Bidding:** Researchers Miller and Drexler propose that computational "objects" should bid for scarce resources like RAM and CPU time. This decentralized approach is intended to manage complexity more effectively than "central planning" by a single operating system.

The Research Agenda of "High-Tech Hayekians"

The "Agorics Project" at George Mason University identified five distinct research areas:

1. **Case Studies:** Analyzing the computer industry using market-process tools.
2. **Hypertext Evolution:** Using systems like Xanadu to create "multidirectional linking" for scholarly criticism, allowing refutations to be permanently attached to original articles to speed the evolution of knowledge.
3. **Programming Practices:** Viewing modularity and specialization as a "division of labor" within software.
4. **AI as Spontaneous Order:** Drawing parallels between Hayek's *The Sensory Order* and neural network research.
5. **Discovery Procedures:** Using computer simulations as "mental experiments" to test economic theories of how money or clearinghouses evolve.

3. The Evolution of Digital Money and Privacy

The sources place heavy emphasis on "Chaumian" digital cash and its potential to revolutionize the global economy by enforcing anonymity.

Characteristics of Digital Cash

Unlike credit cards or checks, true digital cash (such as Ecash or "Magic Money") allows for anonymous transactions.

- **The Double-Spending Problem:** Because digital data is easily copied, systems must use "blinding" and cryptographic signatures to ensure a number cannot be spent twice while still maintaining the user's anonymity from the bank.
- **Privacy vs. The "Four Horsemen":** While governments cite the "Four Horsemen of the Infocalypse" (drug dealers, terrorists, money launderers, and child pornographers) as reasons to ban digital cash, proponents argue that such bans only harm legitimate users while criminals will use the technology anyway.

Economic and Political Implications

- **Offshore Banking:** Desktop electronic funds transfer is predicted to bring offshore banking to the "small player," allowing individuals to escape domestic taxes, reserve requirements, and "monetary supranationalism."
- **Erosion of State Power:** Anonymous digital cash is expected to sharply reduce the state's power to tax and control international money flows.
- **Reputation Markets:** In the absence of state regulation, the digital economy would rely on "escrow agents" and reputation-based systems to prevent fraud.

4. Intelligence and Spontaneous Order

The documents distinguish between traditional AI and "Emergent AI," aligning the latter with Hayekian philosophy.

- **Critique of Rationalism:** Traditional AI (Expert Systems) is criticized for trying to "mirror the world with a formal theory," a method that has failed to produce true expertise.
- **Emergent Intelligence:** Intelligence is viewed as an emergent property of decentralized, independent elements (neurons or rules) interacting. This mirrors the market process, where the "social intelligence" of the system exceeds the intelligence of any individual part.
- **Neural Networks:** These systems are designed to learn rather than follow fixed rules, reflecting the biological processes described in Hayek's psychological works.

5. Future Forecasts and Transhumanist Themes

The Extropy Institute documents provide a timeline of expected breakthroughs, characterized by "Dynamic Optimism."

Technological Projections (1995–2300)

A panel of prognosticators (including Eric Drexler, Nick Szabo, and Mark Miller) provided estimates for several milestones: | Event | Earliest Forecast | Latest Forecast / Never || ----- | ----- | ----- || **Anonymous Electronic Cash (1M+ Users)** | 1990s | 2010 || **Biotech Cures (Aging/Cancer)** | 2010 | 2040 || **Human-Level AI** | 2015 | 2050 || **Nanotech Factories** | 2010 | 2050 || **Uploaded Minds** | 2025 | 2150 || **Privatized Law Enforcement** | 1990s | 2020 || **First Person on Mars** | 2010 | 2030 |

Key Definitions

- **Transhumanism:** A philosophy that seeks to accelerate the evolution of intelligent life beyond human form and limitations using science and technology.
- **Extropy:** A measure of intelligence, information, energy, and growth.
- **The Singularity:** A point of rapid technological onset, often associated with molecular assembly capabilities. Mark Miller predicts this center of gravity to occur between 2005 and 2035.

6. Historical and Intellectual Context

Core Intellectual Influences

- **F.A. Hayek:** His theories on spontaneous order, property rights, and the limitations of central planning provide the theoretical backbone for Agoric systems and Emergent AI.
- **Don Lavoie:** Facilitated the "Agorics Project" at George Mason University, bridging the gap between Austrian economics and computer science.
- **David Chaum:** Invented the "Chaumian" digital cash system used in early anonymous payment implementations.

Concluding Quote

"Those societies which practice Freedom of Speech have prospered to a degree that vastly exceeds the success of any civilization that suppressed free speech... There is only one way to ensure we will always get the maximum of useful criticism, and that is to treat freedom of speech AS IF it were sacred, fundamental, holy, inalienable!" — *David Brin, 1995*